

Mouse Framework - Project Report

AI Task Delegation Framework for Human-AI Collaboration

Executive Summary

Mouse Framework is an interactive web application designed to teach users the principles of effective human-AI collaboration. The application focuses on helping users understand when to delegate tasks to AI agents and when human judgment is required, with emphasis on stakes-based decision making, reversibility, and maintaining user control. **Live Repository:** <https://github.com/growingpenguin/mouse-framework>

Problem Statement

As AI agents become increasingly capable of executing complex tasks, users face a critical challenge: determining which tasks can be safely delegated to AI and which require human oversight. Poor delegation decisions can lead to:

- Irreversible actions taken without proper review
- Sensitive data exposed to automated systems
- Legal, financial, or security risks from AI-made decisions
- Loss of user control over critical processes

Solution: The Mouse Framework

The Mouse Framework provides an educational, interactive interface that:

1. **Visualizes task decomposition** - Breaks complex requests into individual task nodes
2. **Classifies stakes levels** - Identifies high-stakes vs low-stakes tasks
3. **Teaches delegation principles** - Shows which tasks are safe to delegate
4. **Emphasizes reversibility** - Demonstrates undo, rollback, and preview controls
5. **Reinforces learning** - Summarizes key takeaways for user retention

Application Flow

The application consists of 5 connected screens:

Screen 1: Task Input & Decomposition

What do you want to do?

Enter a high-level request and the AI will decompose it into tasks

Handle this patient report and notify the legal team

▶ Try Demo (No API Key)

✦ Decompose with Gemini

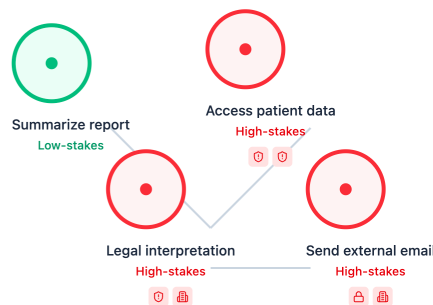
"What do you want to do?" Users enter a high-level request (e.g., "Handle this patient report and notify the legal team"). The system decomposes this into individual task nodes displayed as a visual framework:

- **Green nodes** = Low-stakes (reversible, informational)
- **Red nodes** = High-stakes (irreversible, sensitive, critical)

Key message: "High-stakes tasks should not be delegated to AI."

Screen 2: Framework View (Editable Process)

Task Framework



● Low-stakes (reversible, informational) ● High-stakes (irreversible, sensitive)

ⓘ High-stakes tasks should not be delegated to AI
Tasks involving sensitive data, irreversible actions, or legal/medical decisions require human oversight.

"Your process framework"

Your process framework

The AI proposes a framework. You control it.



Users can interact with the task framework:

- Click nodes to select them
- Remove tasks from the workflow
- Edit descriptions
- Merge or split tasks

Key message: "The AI proposes a framework. You control it."

Screen 3: Delegation & Stakes Rules

"Who should handle each part?"

Who should handle each part?

Delegation rules based on task stakes and reversibility

● Summarize report Low-stakes

✓ Delegate to AI



Enabled

● Access patient data High-stakes

🔒 Security impact

⚖️ Legal/medical impact



Disabled

✗ Human decision required

● Medical interpretation High-stakes

⚖️ Legal/medical impact



Disabled

✗ Human decision required



Delegation Rule

AI can **assist** high-stakes tasks by providing information, analysis, and recommendations, but should not **decide** them. Final decisions on irreversible, sensitive, or critical actions must remain with humans.

1

Tasks delegated to AI

3

Tasks requiring human control

[Continue to Reversibility →](#)

Each task displays its delegation status:

- Low-stakes → "Delegate to AI" (enabled)
- High-stakes → "Human decision required" (disabled)

Risk types are clearly shown:

- Irreversible action
- Legal/medical/security impact
- Company-wide risk

Key message: "AI can assist high-stakes tasks, but should not decide them."

Screen 4: Reversibility & Checkpoints

"Control & safety"

Control & safety

Every action is reversible and protected by checkpoints

Reversibility Controls



Undo

Reverse any action immediately after execution



Rollback

Return to any previous state in the workflow



Preview before commit

Review all changes before they take effect

Mandatory Checkpoints



Sending data

Review recipients and content before sending

- Mandatory approval

✔ Human approval required



Making payments

Confirm amount, recipient, and payment method

- Mandatory approval

✔ Human approval required



Changing security settings

Verify changes to access controls and permissions

- Mandatory approval

✔ Human approval required

Example Workflow

1	AI drafts email Automated execution	
2	Preview & edit draft User review checkpoint	
3	Send email Requires explicit approval	

Delegation never removes your control

You maintain full authority to review, modify, or cancel any action at any stage

Demonstrates safety controls:

- **Undo** - Reverse any action immediately
- **Rollback** - Return to any previous state

- **Preview** - Review changes before commit

Mandatory checkpoints before:

- Sending data externally
- Making payments
- Changing security settings

Key message: "Delegation never removes your control."

Screen 5: Learning & User Improvement

"What you learned"

What you learned

Understanding when to delegate and when to decide



Safely Delegated

- Summarize report

✓ Low-stakes, reversible, informational tasks



Required Human Control

- Access patient data
- Medical interpretation
- Notify stakeholders

⚠ High-stakes, irreversible, sensitive tasks

Key Takeaways

Principles for effective human-AI collaboration

1

Low-stakes, reversible tasks are safe to delegate

Tasks that can be undone, involve no sensitive data, and have minimal consequences can be fully automated.

2

High-stakes, irreversible tasks require human judgment

Decisions with legal, financial, security, or ethical implications must include human oversight and final approval.

3

AI assists, but you maintain control

The AI helps you build and execute your process, but you always retain the ability to review, modify, and override decisions.

Core Principle

The AI helps you **improve your process**, not replace your decisions. Think of it as a framework builder that enhances your workflow while keeping you in control of critical choices.



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Collaboration Summary

4

Total tasks

25%

Automated

75%

Human-guided

Try Another Workflow

Summarizes the session:

- Tasks safely delegated to AI
- Tasks requiring human control
- Collaboration statistics (% automated vs human-guided)

Key takeaways:

1. Low-stakes, reversible tasks are safe to delegate
2. High-stakes, irreversible tasks require human judgment
3. AI assists, but you maintain control

Core principle: "The AI helps you improve your process, not replace your decisions."

Technical Implementation

Technology Stack

Technology	Purpose
React 18	UI framework
TypeScript	Type safety
Vite	Build tool & dev server
Tailwind CSS 4	Styling
Google Gemini AI	Task decomposition
Lucide React	Icons
Radix UI	Accessible UI primitives

AI Integration

The application integrates Google Gemini AI (gemini-1.5-flash model) for intelligent task decomposition:

- **Free tier usage:** 15 requests/minute, 1,500 requests/day, 1M tokens/month
- **Smart fallback:** Works without API key using keyword-based task generation
- **Demo mode:** Pre-built example for immediate exploration

Project Structure

```
mouse-framework/
├── src/
│   ├── components/
│   │   ├── TaskNode.tsx          # Visual task nodes
│   │   ├── TaskInput.tsx        # Screen 1
│   │   ├── FrameworkView.tsx    # Screen 2
│   │   ├── DelegationRules.tsx  # Screen 3
│   │   ├── ReversibilityCheckpoints.tsx # Screen 4
│   │   └── LearningOutcome.tsx  # Screen 5
│   ├── lib/
│   │   └── gemini.ts            # AI integration
│   └── styles/
├── .env.example                 # API key template
├── package.json
└── README.md
```

Design Principles

Principle	Implementation
Visual Stakes	Green (safe) vs Red (danger) color coding
User Control	Editable framework, not fixed rules
Reversibility	Undo, rollback, preview at every step
Education	Teaching users over time, not just blocking
Enterprise-grade	Professional, calm, instructional UI
Accessibility	Keyboard navigation, clear visual hierarchy

Installation & Usage

Quick Start (Demo Mode)

```
git clone https://github.com/growingpenguin/mouse-framework.git
```



```
cd mouse-framework  
  
npm install  
  
npm run dev
```

Open <http://localhost:5173> and click **"Try Demo (No API Key)"** to explore.

Full Setup (With AI)

```
git clone https://github.com/growingpenguin/mouse-framework.git  
  
cd mouse-framework  
  
npm install  
  
cp .env.example .env  
  
# Add your Gemini API key to .env  
  
npm run dev
```

Get a free API key at: <https://aistudio.google.com/app/apikey>

Key Features

1. **No API Key Required** - Demo mode allows instant exploration
2. **Gemini AI Integration** - Intelligent task decomposition (free tier)
3. **Interactive Visualization** - Node-based task framework
4. **Stakes Classification** - Automatic high/low stakes detection
5. **Risk Type Identification** - Security, legal, irreversibility flags
6. **Educational Flow** - 5 screens teaching delegation principles
7. **Responsive Design** - Works on desktop browsers
8. **Modern UI** - Clean, professional enterprise aesthetic

Learning Outcomes

Users who complete the Mouse Framework experience will understand:

1. **When to delegate to AI:**
 - Reversible actions
 - Informational tasks
 - Low-consequence decisions
 - Internal drafts and analysis
2. **When human judgment is required:**

- Irreversible actions
 - Sensitive data access
 - Legal/medical/financial decisions
 - Security-impacting changes
 - External communications
3. **How to maintain control:**
- Review before commit
 - Undo capabilities
 - Checkpoint approvals
 - Editable frameworks

Future Enhancements

Potential areas for expansion:

- User authentication and saved workflows
- Custom task templates
- Team collaboration features
- Integration with real AI agent systems
- Analytics and learning progress tracking
- Mobile-responsive design

Conclusion

The Mouse Framework demonstrates that effective human-AI collaboration requires clear boundaries, visual feedback, and educational reinforcement. By treating AI as a "framework builder" rather than a decision-maker, users maintain control while benefiting from AI assistance. The application successfully teaches the core principle: **AI helps improve your process, not replace your decisions.**